

Table 1: CP10 Metric - Best values for T-ECD Small Retail highlighted in bold

ϕ	Threshold	CP10			
		$\hat{s}_{u,i}$	$\hat{s}_{u,i}^F$	$\hat{s}_{u,i}^{FI}$	$\hat{s}_{u,i}^{FU}$
0.1	0.5	0.47384	0.50980	0.48111	0.53335
0.1	0.6	0.47162	0.50993	0.48164	0.46747
0.1	0.7	0.47505	0.50765	0.48165	0.55279
0.2	0.5	0.46927	0.51031	0.48056	0.55476
0.2	0.6	0.46943	0.51096	0.48600	0.48459
0.2	0.7	0.47396	0.50684	0.48083	0.53157
0.3	0.5	0.46885	0.51093	0.48306	0.46643
0.3	0.6	0.46840	0.51152	0.48039	0.51430
0.3	0.7	0.47591	0.50779	0.48444	0.46902

Table 2: CP100 Metric - Best values for T-ECD Small Retail highlighted in bold

ϕ	Threshold	CP100			
		$\hat{s}_{u,i}$	$\hat{s}_{u,i}^F$	$\hat{s}_{u,i}^{FI}$	$\hat{s}_{u,i}^{FU}$
0.1	0.5	0.34851	0.38483	0.35738	0.39017
0.1	0.6	0.34830	0.38497	0.35851	0.34892
0.1	0.7	0.34881	0.38262	0.35870	0.39762
0.2	0.5	0.34762	0.38542	0.35866	0.39826
0.2	0.6	0.34812	0.38538	0.36341	0.35493
0.2	0.7	0.34891	0.38278	0.35773	0.38826
0.3	0.5	0.34810	0.38596	0.36039	0.34820
0.3	0.6	0.34740	0.38615	0.35941	0.38023
0.3	0.7	0.34907	0.38300	0.36212	0.34859

Table 3: CDCG Metric - Best values for T-ECD Small Retail highlighted in bold

ϕ	Threshold	CDCG			
		$\hat{s}_{u,i}$	$\hat{s}_{u,i}^F$	$\hat{s}_{u,i}^{FI}$	$\hat{s}_{u,i}^{FU}$
0.1	0.5	9.60196	10.00719	9.68985	10.08335
0.1	0.6	9.59458	10.00588	9.70488	9.59080
0.1	0.7	9.60657	9.98735	9.71841	10.17846
0.2	0.5	9.58234	10.00926	9.70525	10.18996
0.2	0.6	9.58722	10.01190	9.75872	9.67585
0.2	0.7	9.60190	9.98725	9.69484	10.06519
0.3	0.5	9.58638	10.01342	9.71756	9.58089
0.3	0.6	9.58439	10.01571	9.70864	9.97747
0.3	0.7	9.60991	9.99080	9.75532	9.58890

Table 4: SA-DIR@10 Metric - Best values for T-ECD Small Retail highlighted in bold

ϕ	Threshold	DIR@10			
		$\hat{s}_{u,i}$	$\hat{s}_{u,i}^F$	$\hat{s}_{u,i}^{FI}$	$\hat{s}_{u,i}^{FU}$
0.1	0.5	-2.27187	-3.07606	-2.48081	-3.38473
0.1	0.6	-2.23905	-3.05943	-2.46139	-2.22781
0.1	0.7	-2.27679	-3.01120	-2.47044	-3.56412
0.2	0.5	-2.22382	-3.08889	-2.47633	-3.61105
0.2	0.6	-2.21704	-3.08080	-2.55280	-2.39261
0.2	0.7	-2.27076	-3.00868	-2.44793	-3.39182
0.3	0.5	-2.20698	-3.09491	-2.51363	-2.21721
0.3	0.6	-2.20416	-3.10097	-2.47807	-3.02719
0.3	0.7	-2.27077	-3.01465	-2.50695	-2.25193

Table 5: RCAU@10 Metric - Best values for T-ECD Small Retail highlighted in bold

ϕ	Threshold	RCAU@10			
		$\hat{s}_{u,i}$	$\hat{s}_{u,i}^F$	$\hat{s}_{u,i}^{FI}$	$\hat{s}_{u,i}^{FU}$
0.1	0.5	0.46462	0.50188	0.47215	0.52628
0.1	0.6	0.46232	0.50202	0.47270	0.45802
0.1	0.7	0.46587	0.49966	0.47271	0.54642
0.2	0.5	0.45989	0.50241	0.47158	0.54847
0.2	0.6	0.46005	0.50309	0.47722	0.47576
0.2	0.7	0.46475	0.49881	0.47186	0.52443
0.3	0.5	0.45945	0.50305	0.47417	0.45694
0.3	0.6	0.45899	0.50367	0.47140	0.50655
0.3	0.7	0.46676	0.49980	0.47560	0.45962

Table 6: NDCG@10 Metric - Best values for T-ECD Small Retail highlighted in bold

ϕ	Threshold	NDCG@10			
		$\hat{s}_{u,i}$	$\hat{s}_{u,i}^F$	$\hat{s}_{u,i}^{FI}$	$\hat{s}_{u,i}^{FU}$
0.1	0.5	0.63576	0.70228	0.65804	0.71067
0.1	0.6	0.63276	0.70118	0.65433	0.66383
0.1	0.7	0.63739	0.70151	0.65931	0.72528
0.2	0.5	0.63232	0.70245	0.65711	0.72589
0.2	0.6	0.63236	0.70086	0.66017	0.67320
0.2	0.7	0.63748	0.70171	0.65590	0.71147
0.3	0.5	0.63057	0.70226	0.65757	0.66244
0.3	0.6	0.63106	0.70288	0.65527	0.69627
0.3	0.7	0.63724	0.70055	0.66127	0.66304

Table 7: Recall@10 Metric - Best values for T-ECD Small Retail highlighted in bold

ϕ	Threshold	Recall@10			
		$\hat{s}_{u,i}$	$\hat{s}_{u,i}^F$	$\hat{s}_{u,i}^{FI}$	$\hat{s}_{u,i}^{FU}$
0.1	0.5	0.30460	0.30969	0.30649	0.30999
0.1	0.6	0.30405	0.30938	0.30552	0.30567
0.1	0.7	0.30415	0.30942	0.30607	0.31063
0.2	0.5	0.30331	0.30879	0.30524	0.31011
0.2	0.6	0.30396	0.30926	0.30632	0.30641
0.2	0.7	0.30448	0.30976	0.30627	0.30987
0.3	0.5	0.30325	0.30895	0.30560	0.30488
0.3	0.6	0.30377	0.30927	0.30577	0.30857
0.3	0.7	0.30458	0.30923	0.30588	0.30522

Table 8: Precision@10 Metric - Best values for T-ECD Small Retail highlighted in bold

ϕ	Threshold	Precision@10			
		$\hat{s}_{u,i}$	$\hat{s}_{u,i}^F$	$\hat{s}_{u,i}^{FI}$	$\hat{s}_{u,i}^{FU}$
0.1	0.5	0.56165	0.62130	0.58143	0.62841
0.1	0.6	0.55993	0.62110	0.57902	0.58820
0.1	0.7	0.56363	0.62093	0.58277	0.64077
0.2	0.5	0.55973	0.62222	0.58165	0.64186
0.2	0.6	0.56007	0.62080	0.58451	0.59620
0.2	0.7	0.56276	0.62034	0.57901	0.62867
0.3	0.5	0.55803	0.62225	0.58164	0.58670
0.3	0.6	0.55878	0.62264	0.57989	0.61679
0.3	0.7	0.56415	0.62098	0.58563	0.58797